COMP 399 - Data Intensive Computing

Phase 2 – Data Modeling

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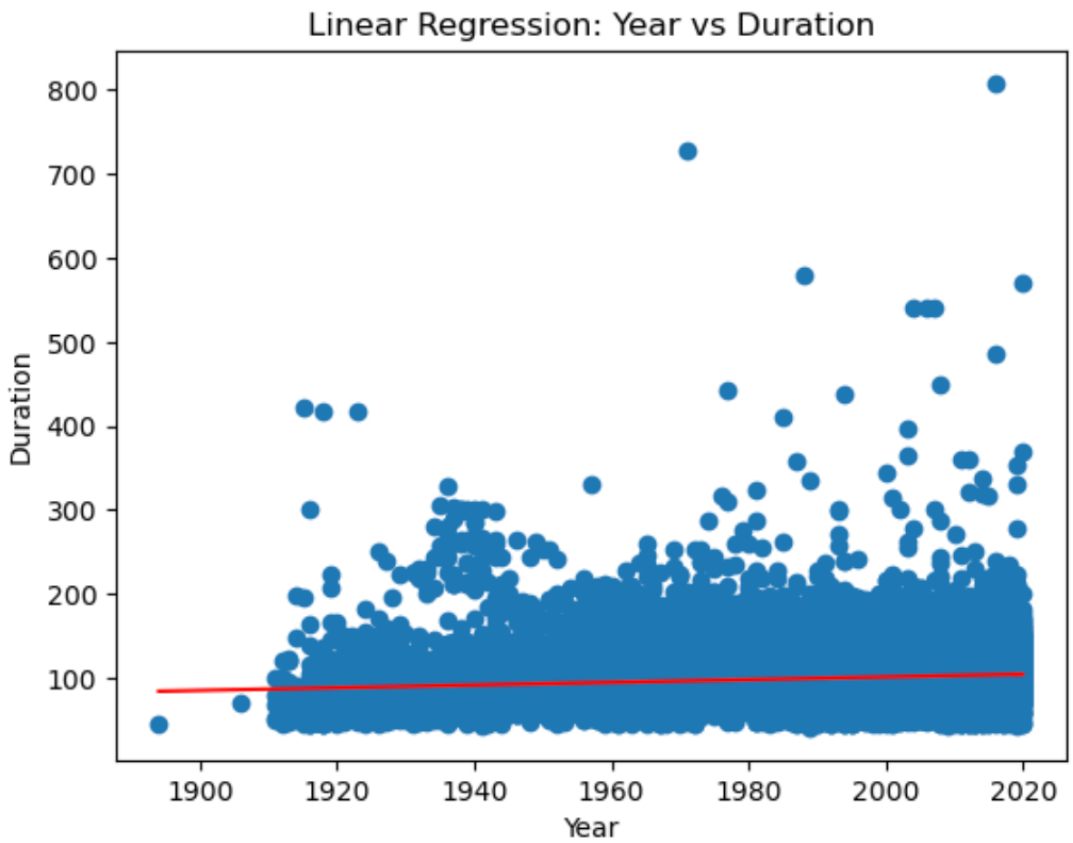
Algorithm 1 - Linear Regression



Linear Regression is a supervised machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables. It works by fitting a straight line that best represents the data.

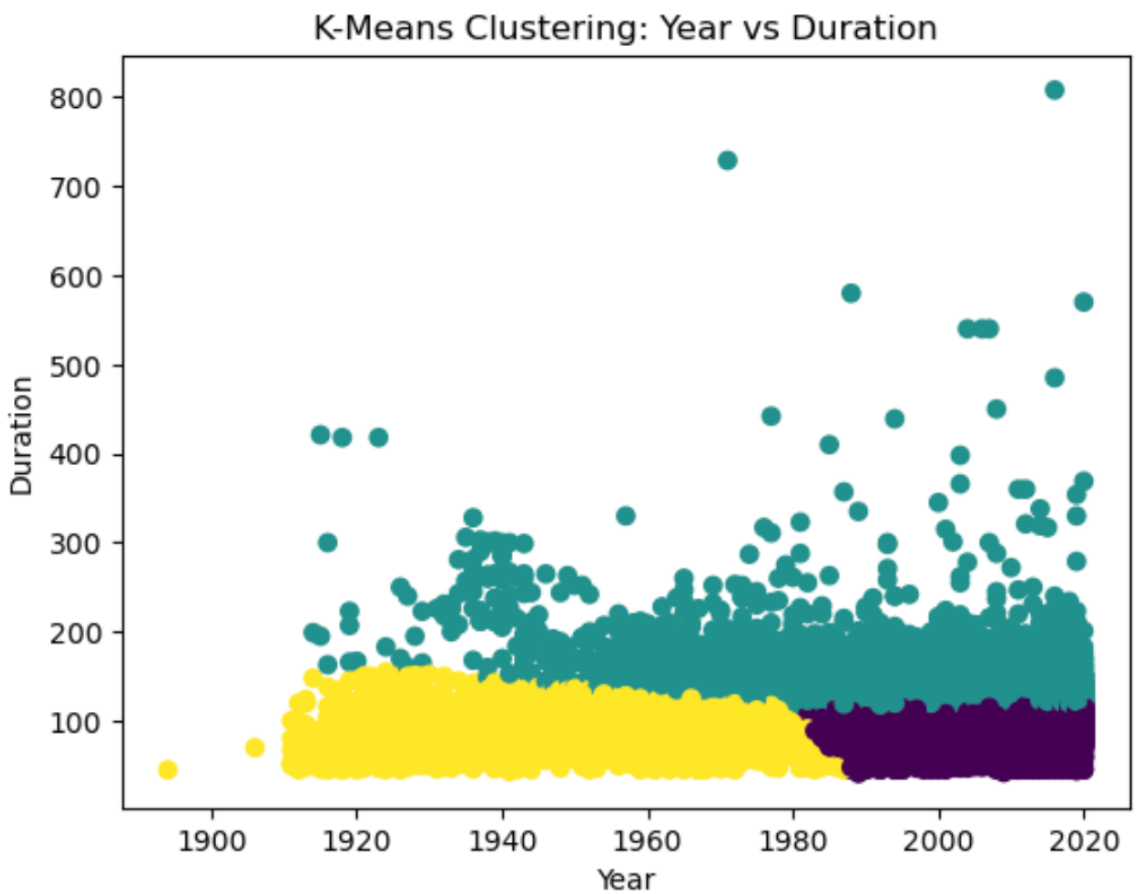
In this project, Linear Regression was used to examine the relationship between movie release year and movie duration. The goal was to understand whether movie lengths have changed over time.

I chose this algorithm because it is simple, easy to interpret, and effective for identifying trends in numerical data. It helps in determining whether there is a positive or negative relationship between variables.

From the results, the regression line shows a slight upward trend. This indicates a small positive relationship between year and duration, suggesting that movie durations have slightly increased over time. However, the data points are widely spread, which means the relationship is weak and other factors may also influence movie duration.

This shows that while Linear Regression is useful for identifying general trends, it may not fully explain the variability in the dataset.

Algorithm 2 - K-Means Clustering



K-Means is an unsupervised machine learning algorithm used to group data into clusters based on similarity. It partitions the dataset into a fixed number of clusters, where each data point belongs to the cluster with the nearest centroid.

In this project, K-Means clustering was applied to group movies based on their year of release and duration. The dataset was divided into three clusters to identify patterns among different types of movies.

I chose this algorithm because it is useful for discovering hidden patterns in data without requiring labeled outputs. It helps in organizing data into meaningful groups.

From the clustering results, the data points are divided into three distinct groups. One cluster mainly represents shorter duration movies, while the other clusters represent medium and longer duration movies. The clusters also show variation across different time periods.

This indicates that K-Means is effective in identifying natural groupings in the dataset. It provides useful insights into how movies can be categorized based on their duration and release year.

References

Scikit-learn Developers. *Linear Regression*. Retrieved from https://scikit-learn.org/stable/modules/linear_model.html

Scikit-learn Developers. *K-Means Clustering*. Retrieved from https://scikit-learn.org/stable/modules/clustering.html